

# Osptek Display

## TFT LCD SPECIFICATION

Model No:

**YDP399B002-V4**

*osptek*<sup>®</sup>

## 1. Introduction

### 1.1 Scope of application

This specification applies to the LCD module that is supplied by Shenzhen Osprey Optoelectronics Technology Co., Ltd.

LCD specification: Dots 400xRGBx960

As to basic specification of the driver IC, refer to the IC (Sitronix: ST7701SN-G5-1) specification and data book.

All material & processing of the LCD module should be Lead Free.

### 1.2 TFT features:

Structure: TFT PANNEL+IC +FPC+BL;

ALL Viewing Type LCD

400 dot-segment and 960 dot-common outputs;

16.7M Color can be selected by software;

White LED back light;

3line SPI or 18bit RGB interface

### 1.3 Applications:

The logo for Osptek, featuring a stylized bird or wing graphic in light blue and purple above the word "osptek" in a lowercase, sans-serif font, followed by a registered trademark symbol (®).

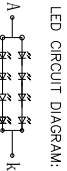
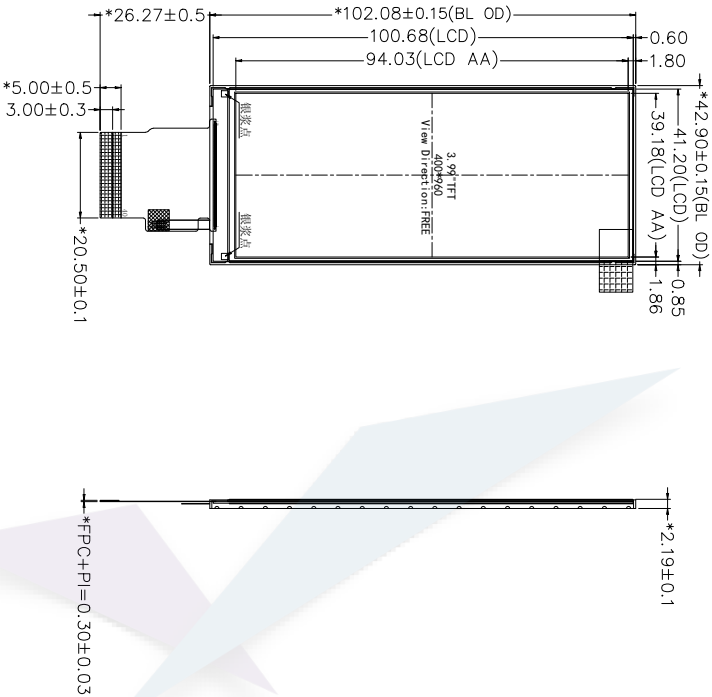
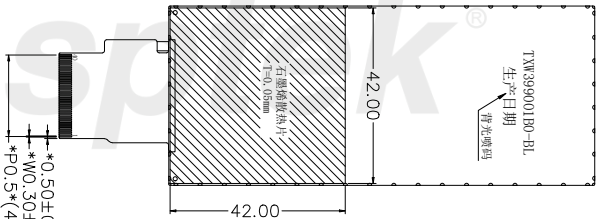
## 2. LCM General specification

| ITEM               | Standard value                   | Unit |
|--------------------|----------------------------------|------|
| LCD Type           | Normally Black                   | --   |
| Drive element      | TFT active matrix                | --   |
| Number of pixels   | 400*3RGB(H)X960(V)               | Dots |
| Pixel arrangement  | RGB Vertical Stripe              | --   |
| Pixel Pitch (W*H)  | 0.108(H) × 0.108(V)              | mm   |
| Active area        | 39.18(H) x 94.03(V)              | mm   |
| Viewing direction  | ALL Viewing                      | --   |
| TFT Driver IC      | ST7701SN-G5-1                    | --   |
| TFT interface      | 3line SPI or 18bit RGB Interface | --   |
| Approx. Weight     | TBD                              | g    |
| Module Size(W*H*T) | 42.90(W) ×102.08(H) ×2.19(T)     | mm   |
| Touch structure    | --                               | --   |
| Touch Driver IC    | --                               | --   |
| Touch Interface    | --                               | --   |

| REV | DESCRIPTION  | DWG. DATE  | DRAWN BY |
|-----|--------------|------------|----------|
| 00  | First Design | 2023.08.29 |          |

|    |              |
|----|--------------|
| 1  | LEDA         |
| 2  | LEDK1        |
| 3  | LEDK2        |
| 4  | GND          |
| 5  | V01 2.8V     |
| 6  | RESET        |
| 7  | NC           |
| 8  | NC           |
| 9  | SDA          |
| 10 | SCK          |
| 11 | CS           |
| 12 | PCLK         |
| 13 | DE           |
| 14 | VS1NC        |
| 15 | HS1NC        |
| 16 | DB0          |
| 17 | DB1          |
| 18 | DB2          |
| 19 | DB3          |
| 20 | DB4          |
| 21 | DB5          |
| 22 | DB6          |
| 23 | DB7          |
| 24 | DB8          |
| 25 | DB9          |
| 26 | DB10         |
| 27 | DB11         |
| 28 | DB12         |
| 29 | DB13         |
| 30 | DB14         |
| 31 | DB15         |
| 32 | DB16         |
| 33 | DB17         |
| 34 | GND          |
| 35 | TP_INT(NC)   |
| 36 | TP_SDA(NC)   |
| 37 | TP_SCL(NC)   |
| 38 | TP_RESET(NC) |
| 39 | TP_VO(NC)    |
| 40 | GND          |

FPC展开出货



- NOTES:
1. DISPLAY TYPE: 3.99", 480\*960 TFT LCD
  2. DISPLAY MODE: transmissive Normally Black
  3. VIEWING DIRECTION: Free
  4. DRIVER IC: ST7701SN-G5-1
  5. LCM (White 9 AVG 1/9) :  
Brightness: 500cd/m<sup>2</sup> (TYP)  
Uniformity: 80% (MIN)
  6. BACK LIGHT: 8 chip white LEDs If=40mA, Vf=22~4V~25.6V
  7. OPERATING TEMP: -20° C TO 70° C, STORAGE TEMP: -30° C TO 80° C
  8. \* Critical Parameter, ( ) ref Parameter, [ ] cpk Parameter  
Unspecified Tolerances: ±0.20mm
  9. SUGGESTION: TP window size unilateral increase 0.3~0.5mm than LCM A.A
  10. REQUIREMENTS ENVIRONMENTAL PROTECTION: RoHS

|               |               |             |            |                    |  |
|---------------|---------------|-------------|------------|--------------------|--|
| 深圳市鱼鹰光电科技有限公司 |               |             |            | Mobile:            |  |
| PART NO.:     | YDP399B002-V4 | MODULE NO.: | 00         | 0755-26810170      |  |
| DRAWN BY:     | LZH           | DATE:       | 2023.08.29 | PROJECTION: 1 OF 1 |  |
| CHECKED BY:   |               | DATE:       |            | SCALE: N.T.S       |  |
| APPROVED BY:  |               | DATE:       |            | SHEET: 1 OF 1      |  |

### 3. Absolute Maximum Rating

| Characteristics           | Symbol           | Min. | Max. | Unit |
|---------------------------|------------------|------|------|------|
| LCM Operating Temperature | T <sub>OPR</sub> | -20  | +70  | °C   |
| LCM Storage Temperature   | T <sub>STG</sub> | -30  | +80  | °C   |
| Humidity                  | RH               | --   | 90   | %    |

### 4. Electrical Characteristics

#### 4.1 TFT DC Characteristics

| Characteristics           | Symbol | Min. | Typ. | Max. | Unit |
|---------------------------|--------|------|------|------|------|
| Supply Voltage for(DC/DC) | VCC    | 2.5  | 2.8  | 3.3  | V    |

#### 4.2 Back-Light Unit Characteristics

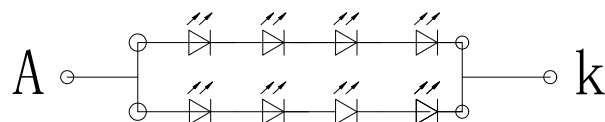
The back-light system is an edge-lighting type with 8 white LEDs. The characteristics of the back-light are shown in the following tables.

| Characteristics     | Symbol         | Min. | Type   | Max. | Unit              | Notes  |
|---------------------|----------------|------|--------|------|-------------------|--------|
| Forward Voltage     | V <sub>F</sub> | 22.4 | --     | 25.6 | V                 | --     |
| Forward current     | I <sub>F</sub> | --   | 40     | --   | mA                | --     |
| Luminance(With LCD) | L <sub>v</sub> | --   | 500    | --   | cd/m <sup>2</sup> | --     |
| LED life time       | N/A            | --   | 30,000 | --   | Hr                | Note 1 |

Note:

- (1) The “LED life time” is defined as the module brightness decrease to 50% of original brightness at I<sub>L</sub>=20mA/LED. The LED life time could be decreased if operating I<sub>L</sub> is larger than 25mA/LED.

Backlight circuit diagram shown in below:



## 5. Module Function Description

| Pin No. | Symbol       | LCM Functional                                                            | Notes |
|---------|--------------|---------------------------------------------------------------------------|-------|
| 1       | LEDA         | Power supply for backlight anode input terminal.                          |       |
| 2       | LEDK1        | Power supply for backlight cathode input terminal.                        |       |
| 3       | LEDK2        | Power supply for backlight cathode input terminal.                        |       |
| 4       | GND          | Power Ground                                                              |       |
| 5       | VCI 2.8V     | Power Supply For LCD.                                                     |       |
| 6       | RESET        | Reset signal input terminal. Active at 'L'.                               |       |
| 7       | NC           | No Connection.                                                            |       |
| 8       | NC           | No Connection.                                                            |       |
| 9       | SDA          | Serial data input/output pin.                                             |       |
| 10      | SCK          | Serial clock signal.                                                      |       |
| 11      | CS           | Chip selection signal.                                                    |       |
| 12      | PCLK         | Dot clock signal for RGB interface operation.                             |       |
| 13      | DE           | Data enable signal for RGB interface operation.                           |       |
| 14      | VSYNC        | Vertical (Frame) synchronizing input signal for RGB interface operation.  |       |
| 15      | HSYNC        | Horizontal (Line) synchronizing input signal for RGB interface operation. |       |
| 16      | DB0          | RGB interface data bus.                                                   |       |
| 17      | DB1          | RGB interface data bus.                                                   |       |
| 18      | DB2          | RGB interface data bus.                                                   |       |
| 19      | DB3          | RGB interface data bus.                                                   |       |
| 20      | DB4          | RGB interface data bus.                                                   |       |
| 21      | DB5          | RGB interface data bus.                                                   |       |
| 22      | DB6          | RGB interface data bus.                                                   |       |
| 23      | DB7          | RGB interface data bus.                                                   |       |
| 24      | DB8          | RGB interface data bus.                                                   |       |
| 25      | DB9          | RGB interface data bus.                                                   |       |
| 26      | DB10         | RGB interface data bus.                                                   |       |
| 27      | DB11         | RGB interface data bus.                                                   |       |
| 28      | DB12         | RGB interface data bus.                                                   |       |
| 29      | DB13         | RGB interface data bus.                                                   |       |
| 30      | DB14         | RGB interface data bus.                                                   |       |
| 31      | DB15         | RGB interface data bus.                                                   |       |
| 32      | DB16         | RGB interface data bus.                                                   |       |
| 33      | DB17         | RGB interface data bus.                                                   |       |
| 34      | GND          | Power Ground                                                              |       |
| 35      | TP-INT(NC)   | No Connection.                                                            |       |
| 36      | TP-SDA(NC)   | No Connection.                                                            |       |
| 37      | TP-SCL(NC)   | No Connection.                                                            |       |
| 38      | TP-RESET(NC) | No Connection.                                                            |       |
| 39      | TP-VCI(NC)   | No Connection.                                                            |       |
| 40      | GND          | Power Ground                                                              |       |

## 6. Timing Characteristics

### POWER ON/OFF SEQUENCE

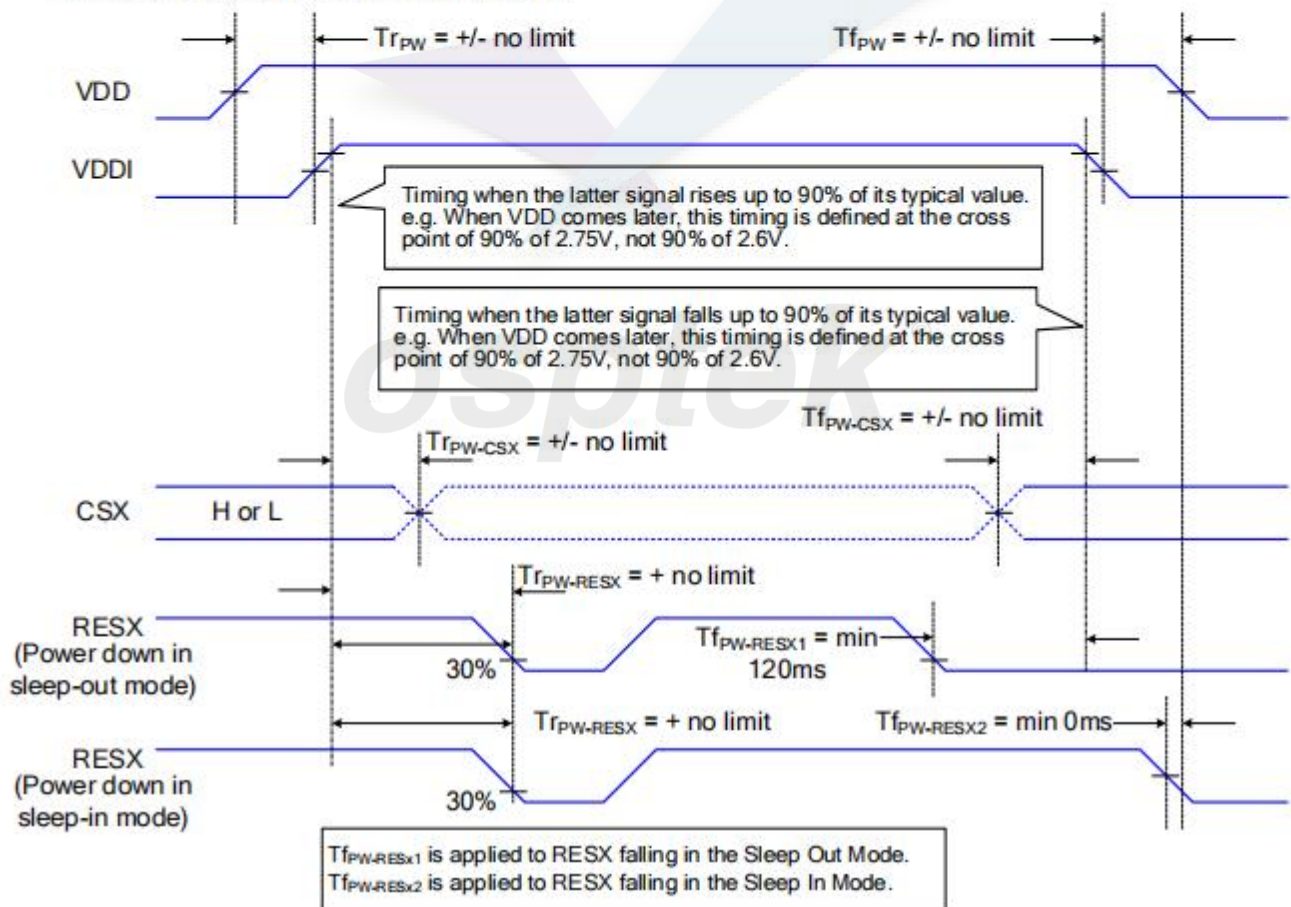
VDDI and VDDA can be applied or powered down in any order. During the Power Off sequence, if the LCD is in the Sleep Out mode, VDDA and VDDI must be powered down with minimum 120msec. If the LCD is in the Sleep In mode, VDDA and VDDI can be powered down with minimum 0msec after the RESX is released.

CSX can be applied at any timing or can be permanently grounded. RESX has high priority over CSX.

Notes:

1. There will be no damage to the ST7701SN if the power sequences are not met.
2. There will be no abnormal visible effects on the display panel during the Power On/Off Sequences.
3. There will be no abnormal visible effects on the display between the end of Power On Sequence and before receiving the Sleep Out command, and also between receiving the Sleep In command and the Power Off Sequence.
4. If the RESX line is not steadily held by the host during the Power On Sequence as defined in Sections 9.1 and 9.2, then it will be necessary to apply the Hardware Reset (RESX) after the completion of the Host Power On Sequence to ensure correct operations. Otherwise, all the functions are not guaranteed.

The power on/off sequence is illustrated below



### Serial Interface Characteristics (3-line serial):

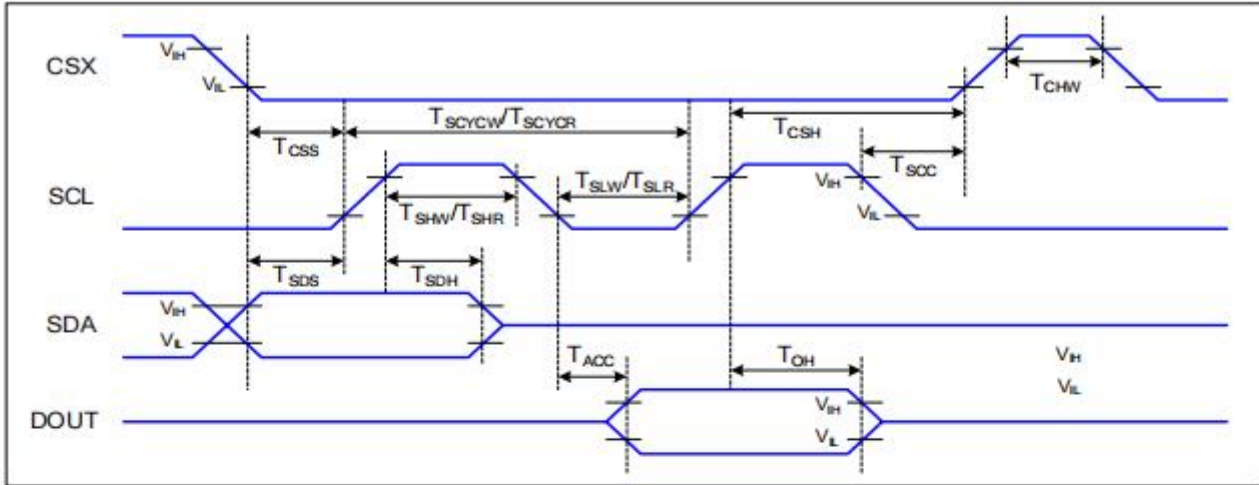


Figure 1 3-line serial Interface Timing Characteristics

VDDI=1.8, VDD=2.8, AGND=DGND=0V, Ta=25 °C

| Signal       | Symbol             | Parameter                      | Min | Max | Unit | Description |
|--------------|--------------------|--------------------------------|-----|-----|------|-------------|
| CSX          | T <sub>CSS</sub>   | Chip select setup time (write) | 15  |     | ns   |             |
|              | T <sub>CSH</sub>   | Chip select hold time (write)  | 15  |     | ns   |             |
|              | T <sub>CSS</sub>   | Chip select setup time (read)  | 60  |     | ns   |             |
|              | T <sub>SCC</sub>   | Chip select hold time (read)   | 60  |     | ns   |             |
|              | T <sub>CHW</sub>   | Chip select "H" pulse width    | 40  |     | ns   |             |
| SCL          | T <sub>SCYCW</sub> | Serial clock cycle (Write)     | 66  |     | ns   |             |
|              | T <sub>SHW</sub>   | SCL "H" pulse width (Write)    | 15  |     | ns   |             |
|              | T <sub>SLW</sub>   | SCL "L" pulse width (Write)    | 15  |     | ns   |             |
|              | T <sub>SCYCR</sub> | Serial clock cycle (Read)      | 150 |     | ns   |             |
|              | T <sub>SHR</sub>   | SCL "H" pulse width (Read)     | 60  |     | ns   |             |
|              | T <sub>SLR</sub>   | SCL "L" pulse width (Read)     | 60  |     | ns   |             |
| SDA<br>(DIN) | T <sub>SDS</sub>   | Data setup time                | 10  |     | ns   |             |
|              | T <sub>SDH</sub>   | Data hold time                 | 10  |     | ns   |             |

Table 4 3-line serial Interface Characteristics

Note : The rising time and falling time (Tr, Tf) of input signal are specified at 15 ns or less. Logic high and low levels are specified as 30% and 70% of VDDI for Input signals.

### RGB Interface Characteristics :

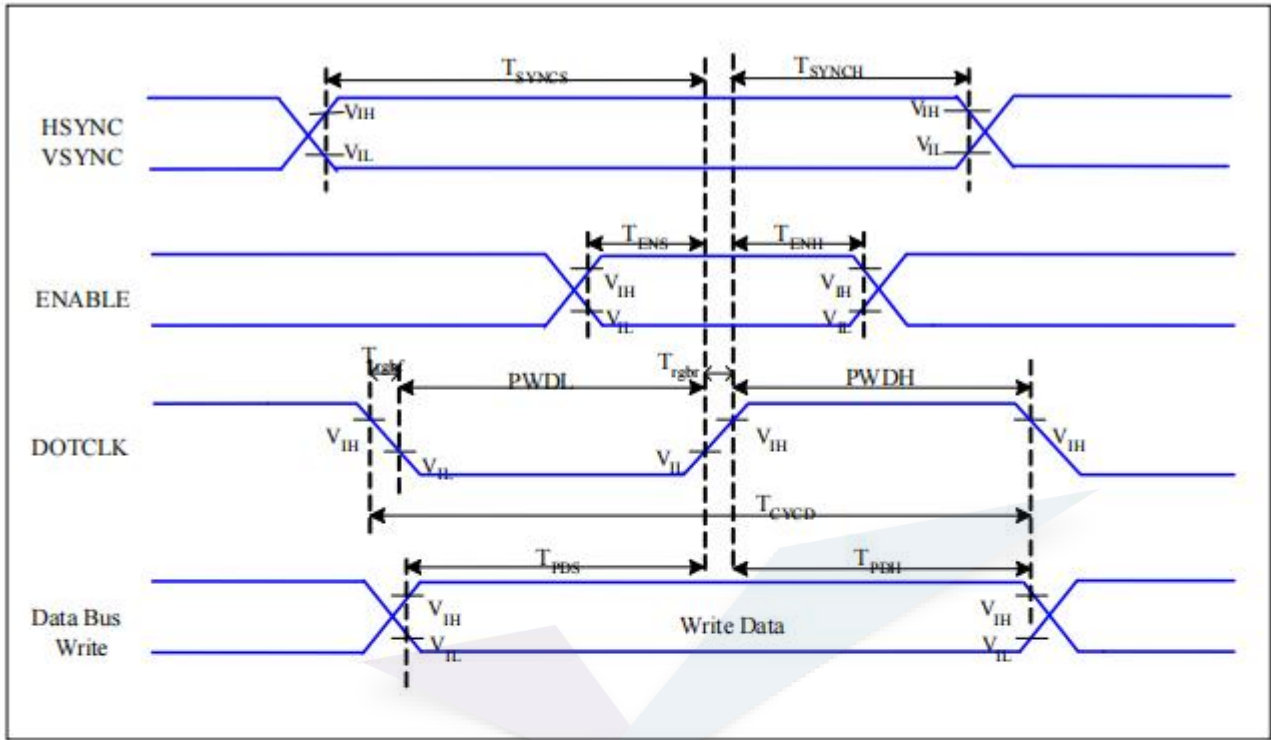


Figure 3 RGB Interface Timing Characteristics

VDDI=1.8, VDD=2.8, AGND=DGND=0V, Ta=25 °C

| Signal       | Symbol               | Parameter                     | MIN | MAX | Unit | Description |
|--------------|----------------------|-------------------------------|-----|-----|------|-------------|
| HSYNC, VSYNC | $T_{SYNCS}$          | VSYNC, HSYNC Setup Time       | 5   | -   | ns   |             |
| ENABLE       | $T_{ENS}$            | Enable Setup Time             | 5   | -   | ns   |             |
|              | $T_{ENH}$            | Enable Hold Time              | 5   | -   | ns   |             |
| DOTCLK       | PWDH                 | DOTCLK High-level Pulse Width | 15  | -   | ns   |             |
|              | PWDL                 | DOTCLK Low-level Pulse Width  | 15  | -   | ns   |             |
|              | $T_{CYCD}$           | DOTCLK Cycle Time             | 33  | -   | ns   |             |
|              | $T_{rghr}, T_{rgfh}$ | DOTCLK Rise/Fall time         | -   | 15  | ns   |             |
| DB           | $T_{PDS}$            | PD Data Setup Time            | 5   | -   | ns   |             |
|              | $T_{PDH}$            | PD Data Hold Time             | 5   | -   | ns   |             |

Table 6 18/16 Bits RGB Interface Timing Characteristics

## Reset Description:

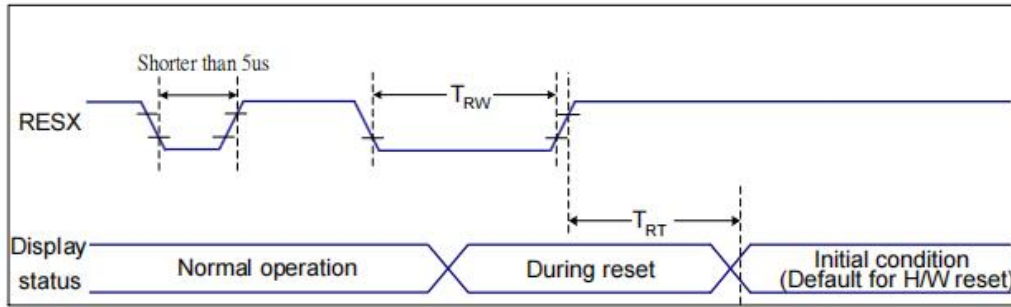


Figure 9 Reset Timing

VDDI=1.8, VDD=2.8, AGND=DGND=0V, Ta=25 °C

| Related Pins | Symbol | Parameter            | MIN | MAX                                 | Unit |
|--------------|--------|----------------------|-----|-------------------------------------|------|
| RESX         | TRW    | Reset pulse duration | 10  | -                                   | us   |
|              | TRT    | Reset cancel         | -   | 5 (Note 1, 5)<br>120 (Note 1, 6, 7) | ms   |

Table 9 Reset Timing

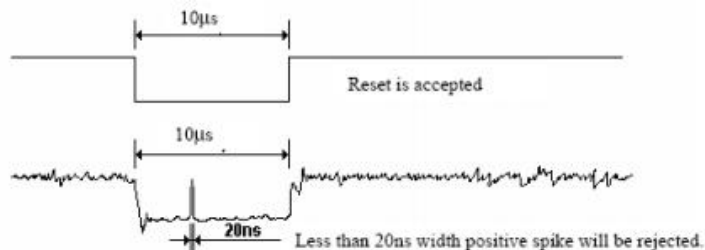
### Notes:

1. The reset cancel includes also required time for loading ID bytes, VCOM setting and other settings from NVM (or similar device) to registers. This loading is done every time when there is HW reset cancel time (tRT) within 5 ms after a rising edge of RESX.
2. Spike due to an electrostatic discharge on RESX line does not cause irregular system reset according to the table below:

| RESX Pulse          | Action         |
|---------------------|----------------|
| Shorter than 5us    | Reset Rejected |
| Longer than 9us     | Reset          |
| Between 5us and 9us | Reset starts   |

3. During the Resetting period, the display will be blanked (The display is entering blanking sequence, which maximum time is 120 ms, when Reset Starts in Sleep Out –mode. The display remains the blank state in Sleep In –mode.) and then return to Default condition for Hardware Reset.

4. Spike Rejection also applies during a valid reset pulse as shown below:



5. When Reset applied during Sleep In Mode.
6. When Reset applied during Sleep Out Mode.
7. It is necessary to wait 5msec after releasing RESX before sending commands. Also Sleep Out command cannot be sent for 120msec.

## 7.Optical Characteristics

### Optical Specification

| Parameter                        |            | Symbol        | Condition                       | Min.  | Typ.  | Max.  | Unit | Remark                                   |
|----------------------------------|------------|---------------|---------------------------------|-------|-------|-------|------|------------------------------------------|
| Viewing Angle range              | Horizontal | $\Theta_3$    | CR > 10                         | 80    | 85    | -     | Deg. | Note 1                                   |
|                                  |            | $\Theta_9$    |                                 | 80    | 85    | -     | Deg. |                                          |
|                                  | Vertical   | $\Theta_{12}$ |                                 | 80    | 85    | -     | Deg. |                                          |
|                                  |            | $\Theta_6$    |                                 | 80    | 85    | -     | Deg. |                                          |
| Contrast ratio                   |            | CR            | $\Theta = 0^\circ$              | 550   | 800   | -     |      | Note 2                                   |
| Transmittance                    |            | Tr.           |                                 | 3.34  | 3.93  | -     | %    | Without APF<br>Note 3                    |
| White Chromaticity               |            | $x_w$         |                                 | 0.277 | 0.292 | 0.307 |      | Note 4<br>CF Glass<br>Base on C<br>Light |
|                                  |            | $y_w$         |                                 | 0.318 | 0.333 | 0.348 |      |                                          |
| Reproduction of color (C light)  | Red        | $R_x$         |                                 |       |       |       |      |                                          |
|                                  |            | $R_y$         |                                 |       |       |       |      |                                          |
|                                  | Green      | $G_x$         |                                 |       |       |       |      |                                          |
|                                  |            | $G_y$         |                                 |       |       |       |      |                                          |
|                                  | Blue       | $B_x$         |                                 |       |       |       |      |                                          |
|                                  |            | $B_y$         |                                 |       |       |       |      |                                          |
| Response Time (Rising + Falling) |            | $T_r + T_f$   | Ta= 25° C<br>$\Theta = 0^\circ$ | -     | 35    | -     | ms   | Note 5                                   |

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Note:

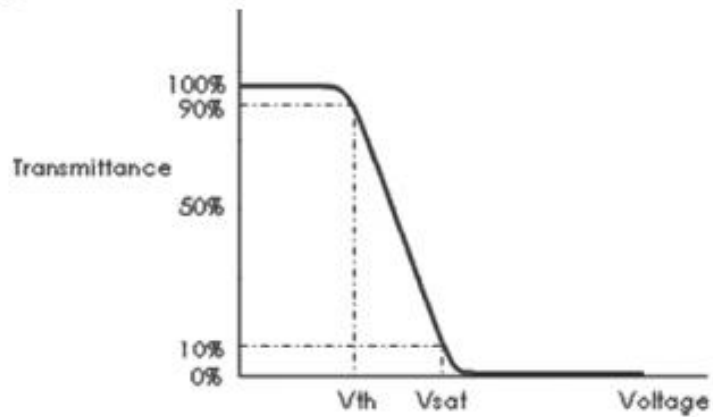
1. Viewing angle is the angle at which the contrast ratio is greater than 10. The viewing angles are determined for the horizontal or 3, 9 o'clock direction and the vertical or 6, 12 o'clock direction with respect to the optical axis which is normal to the LCD surface (see FIGURE 1).
2. Contrast measurements shall be made at viewing angle of  $\Theta = 0$  and at the center of the LCD surface. Luminance shall be measured with all pixels in the view field set first to white, then to the dark (black) state. (see FIGURE 1)  
1) Luminance Contrast Ratio (CR) is defined mathematically.

$$CR = \frac{\text{Luminance when displaying a white raster}}{\text{Luminance when displaying a black raster}}$$

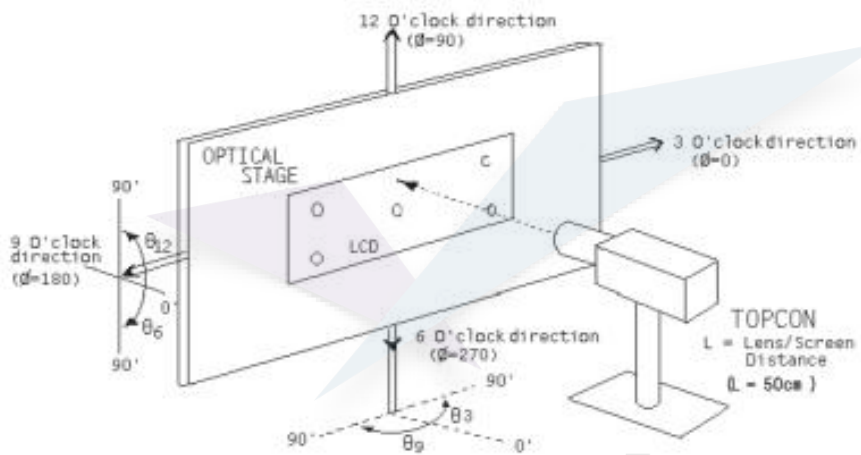
3. Transmittance is the Value with Polarizer
4. The color chromaticity coordinates specified in Table 5 shall be calculated from the spectral data measured with all pixels first in red, green, blue and white. Measurements shall be made at the center of the panel.
5. The electro-optical response time measurements shall be made as FIGURE 3 by switching the "data" input signal ON and OFF. The times needed for the luminance to change from 10% to 90% is  $T_r$ , and 90% to 10% is  $T_d$ .

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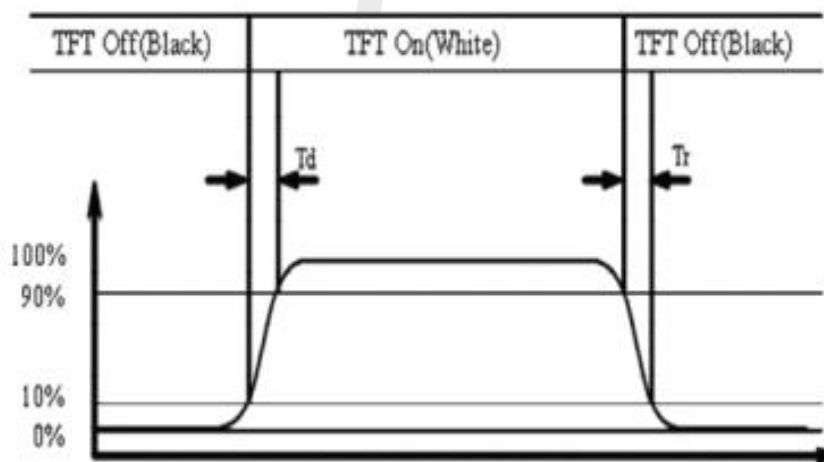
**Figure 1. The Definition of  $V_{th}$  &  $V_{sat}$**



**Figure 2. Measurement Set Up**



**Figure 3. Response Time Testing**



## 8. Reliability Test Item

| No. | Test Item                              | Test Condition                                          | Notes                                                                                                                                                                                                                                   |
|-----|----------------------------------------|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | High Temp. Storage                     | +80°C / 96H                                             | 1. Functional test is OK.<br>Missing Segment, short,<br>unclear segment<br>non-display, display abnormally<br>and liquid crystal leakage<br>un-allowed.<br>2. No low<br>temperature bubbles, end seal<br>loose and fall, frame rainbow. |
| 2   | Low Temp. Storage                      | -30°C / 96H                                             |                                                                                                                                                                                                                                         |
| 3   | High Temp. Operating                   | +70°C / 96H                                             |                                                                                                                                                                                                                                         |
| 4   | Low Temp. Operating                    | -20°C / 96H                                             |                                                                                                                                                                                                                                         |
| 5   | High Temperature<br>/ Humidity storage | 50°C x 90%RH / 96H                                      |                                                                                                                                                                                                                                         |
| 6   | Thermal and cold shock                 | Static state, -20°C (30min) ~60°C<br>(30min), 10 cycles |                                                                                                                                                                                                                                         |

## 9. Packing Method----TBD

- END -

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